Learning Goals

In this chapter, you'll learn…

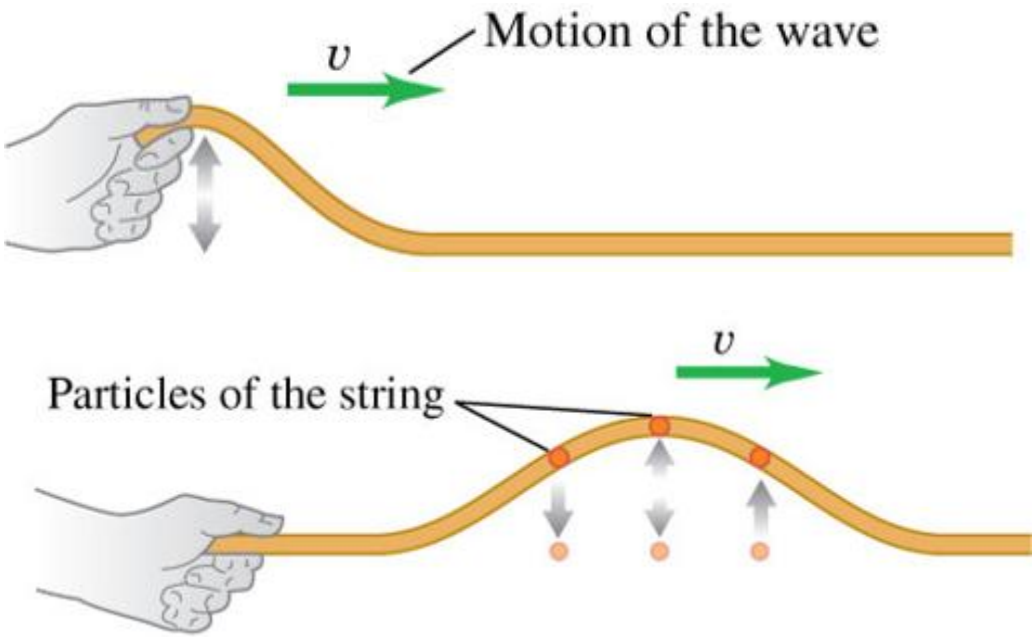
- how to use the relationship among speed, frequency, and wavelength for a periodic wave.
- how to calculate the speed of waves on a rope or string.
- what happens when mechanical waves overlap and interfere.
- the properties of standing waves on a string, and how to analyze these waves.
- how stringed instruments produce sounds of specific frequencies.

Introduction

- Earthquake waves carry enormous power as they travel through the earth.
- Other types of mechanical waves, such as sound waves or the vibration of the strings of a piano, carry far less energy.
- Overlapping waves interfere, which helps us understand musical instruments.

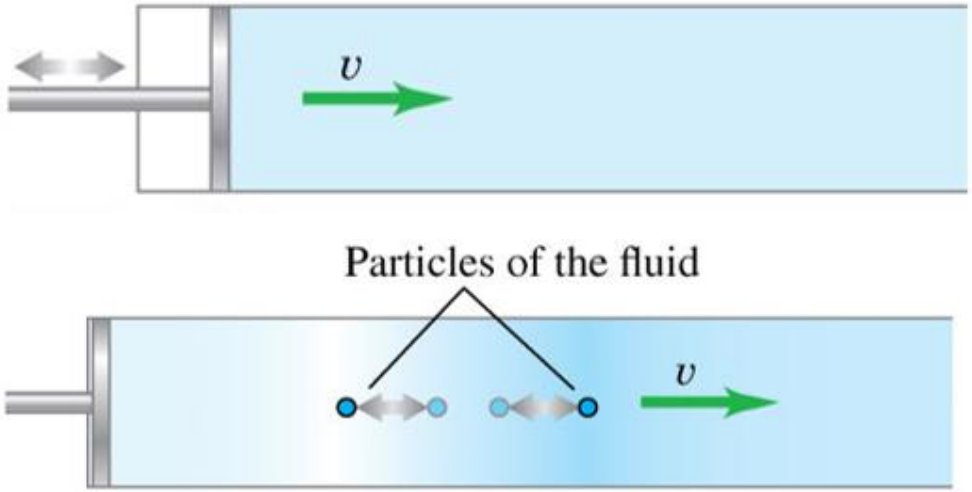
Types of Mechanical Waves (1 of 3)

- A wave on a string is a type of mechanical wave.
- The hand moves the string up and then returns, producing a transverse wave that moves to the right.



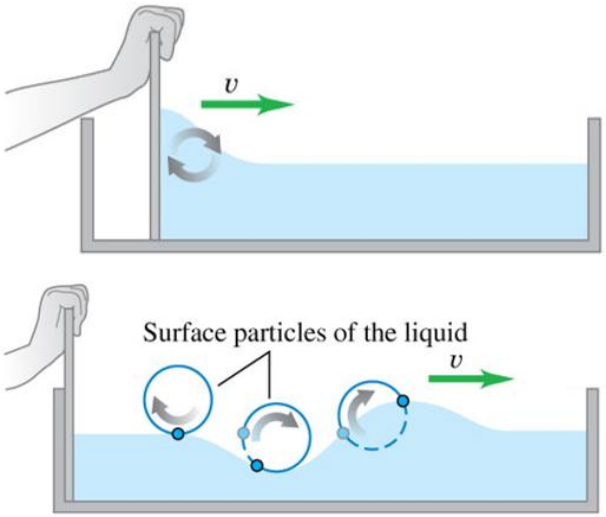
Types of Mechanical Waves (2 of 3)

- A pressure wave in a fluid is a type of mechanical wave.
- The piston moves to the right, compressing the gas or liquid, and then returns, producing a longitudinal wave that moves to the right.



Types of Mechanical Waves (3 of 3)

- A surface wave on a liquid is a type of mechanical wave.
- The board moves to the right and then returns, producing a combination of longitudinal and transverse waves.



Mechanical Waves

- “Doing the wave” at a sports stadium is an example of a mechanical wave.
- The **disturbance** propagates through the crowd, but there is no transport of matter.
- None of the spectators moves from one seat to another.

Periodic Waves (1 of 2)

- For a **periodic wave**, each particle of the medium undergoes periodic motion.
- The **wavelength** λ of a periodic wave is the length of one complete wave pattern.
- The **speed** of any periodic wave of frequency f is:

For a **periodic wave:**

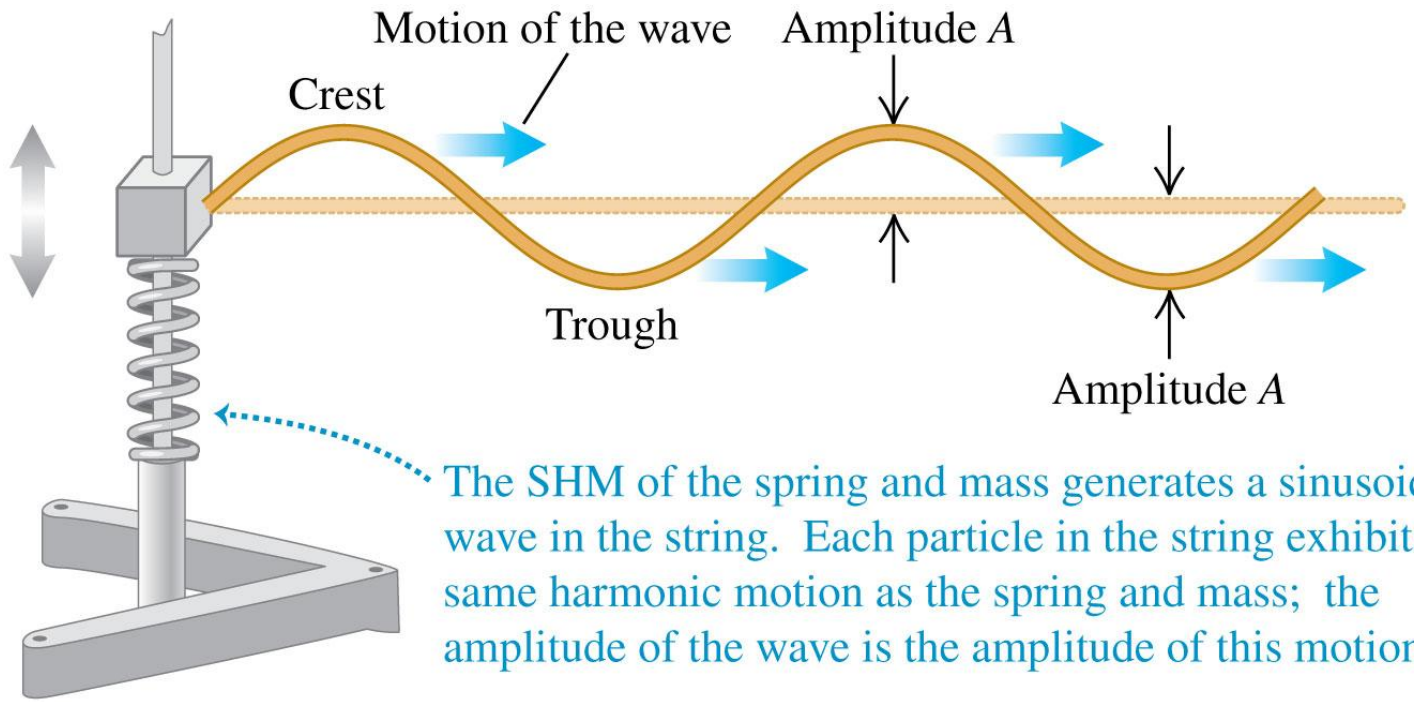
$$\underbrace{v}_{\text{Wave speed}} = \underbrace{\lambda}_{\text{Wavelength}} \underbrace{f}_{\text{Frequency}}$$

- Video Tutor Solution: Example 15.1

Ex. 1 The speed of sound in air at 20°C is 344 m/s. (a) What is the wavelength of a sound wave with a frequency of 784 Hz, corresponding to the note G_5 on a piano, and how many milliseconds does each vibration take? (b) What is the wavelength of a sound wave one octave higher than the note in part (a).

Periodic Transverse Waves

- A mass attached to a spring undergoes simple harmonic motion, producing a sinusoidal wave that travels to the right on the string.



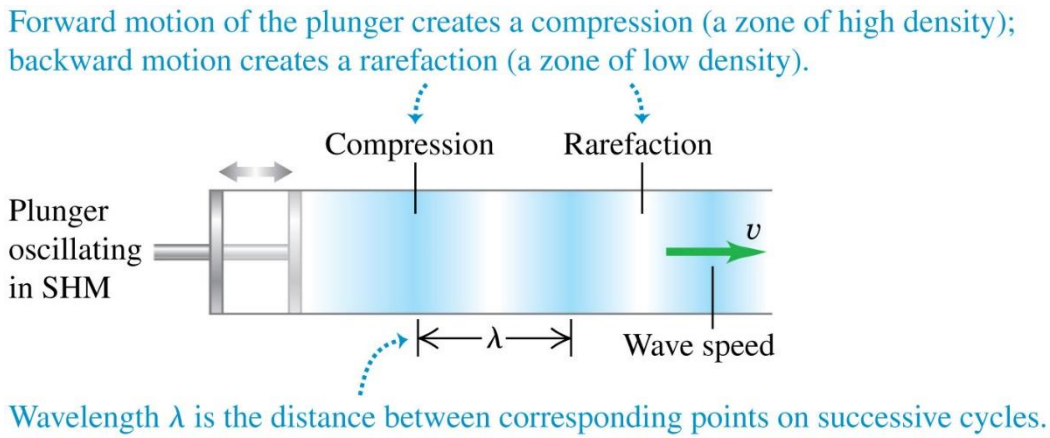
The SHM of the spring and mass generates a sinusoidal wave in the string. Each particle in the string exhibits the same harmonic motion as the spring and mass; the amplitude of the wave is the amplitude of this motion.

Periodic Waves (2 of 2)

- A series of drops falling into water produces a periodic wave that spreads radially outward.
- The wave crests and troughs are concentric circles.
- The wavelength λ is the radial distance between adjacent crests or adjacent troughs.

Periodic Longitudinal Waves

- Consider a long tube filled with a fluid, with a piston at the left end.
- If we push the piston in, we compress the fluid near the piston, and this region then pushes against the neighboring region of fluid, and so on, and a wave pulse moves along the tube.



Mathematical Description of a Wave

- The wave function for a sinusoidal wave moving in the +*x*-direction is given by Equation (15.7):

Wave function for a **sinusoidal wave propagating in +*x*-direction**

Amplitude, Position, Time

$$y(x, t) = A\cos(kx - \omega t)$$

Wave number $= 2\pi/\lambda$, Angular frequency $= 2\pi f$

- In this function, *y* is the displacement of a particle at time *t* and position *x*.
- The quantity *A* is the amplitude of the wave.
- The quantity *k* is called the **wave number**, and is defined as $k = \dfrac{2\pi}{\lambda}$.
- The quantity ω is called the **angular frequency**, and is defined as $\omega = 2\pi f = \dfrac{2\pi}{T}$, where *T* is the period.

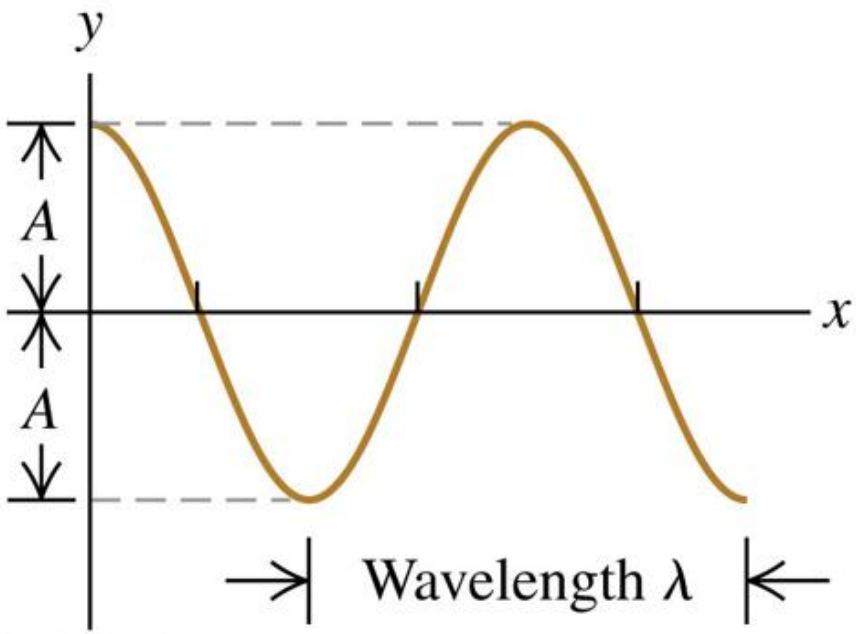
Ex. 8 A certain transverse wave is described by

$$y(x,t) = (6.50\,\mathrm{mm})\cos 2\pi\left(\frac{x}{28.0\,\mathrm{cm}} - \frac{t}{0.0360\,\mathrm{s}}\right)$$

Determine the wave's a) amplitude; b) wavelength; c) frequency; d) speed of propagation; e) direction of propagation.

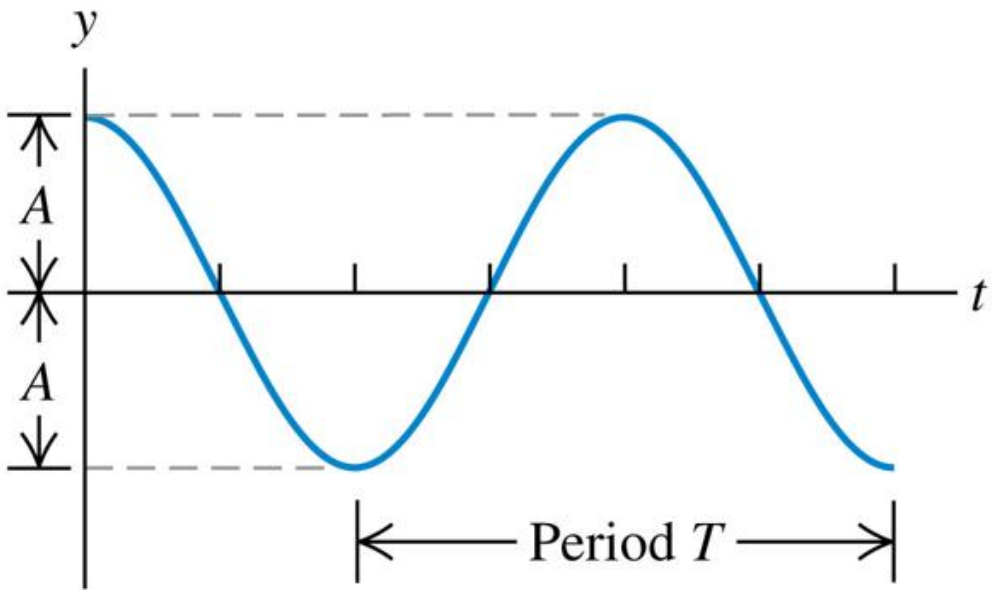
Graphing the Wave Function (1 of 2)

If we use Eq. (15.7) to plot y as a function of x for time $t = 0$, the curve shows the *shape* of the string at $t = 0$.

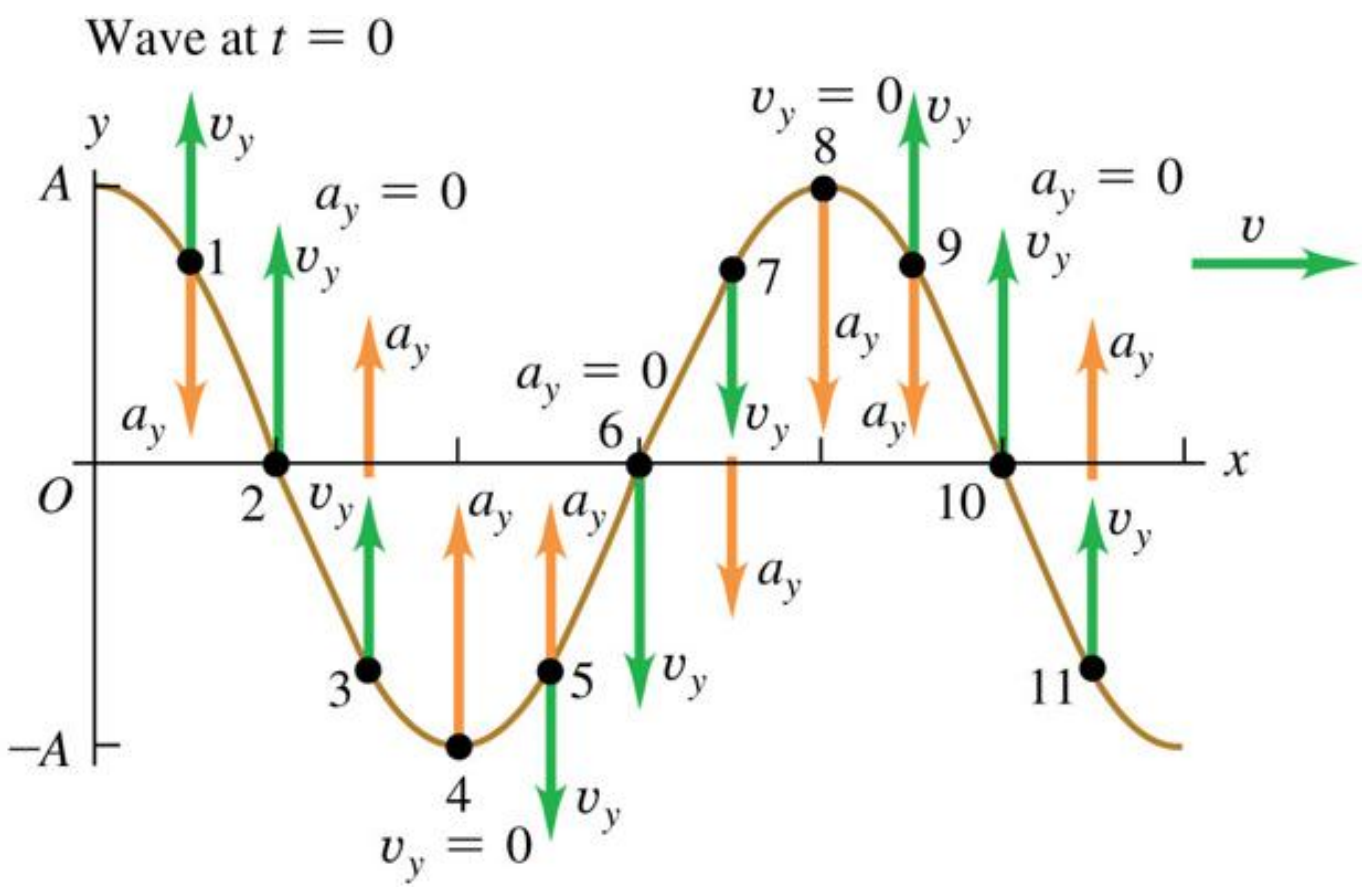


Graphing the Wave Function (2 of 2)

If we use Eq. (15.7) to plot y as a function of t for position $x = 0$, the curve shows the *displacement* y of the particle at $x = 0$ as a function of time.

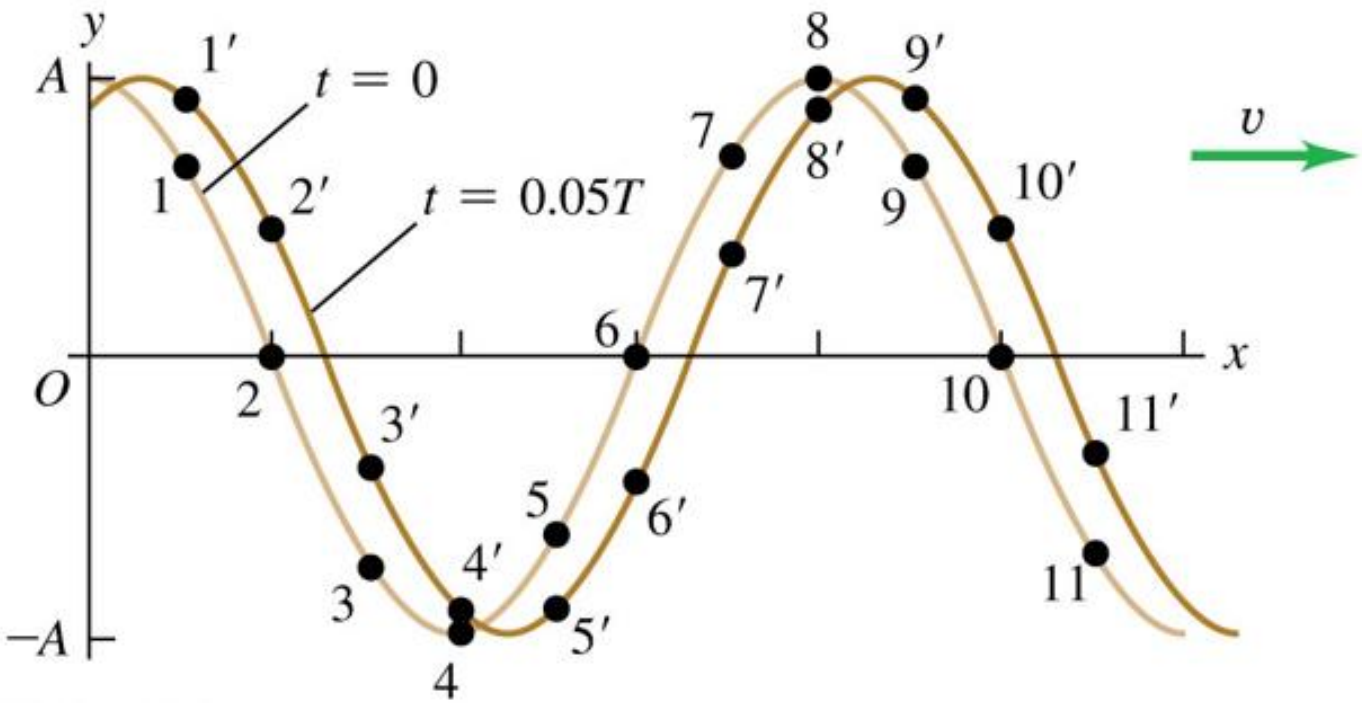


Particle Velocity and Acceleration in a Sinusoidal Wave (1 of 2)



Particle Velocity and Acceleration in a Sinusoidal Wave (2 of 2)

The same wave at $t = 0$ and $t = 0.05T$



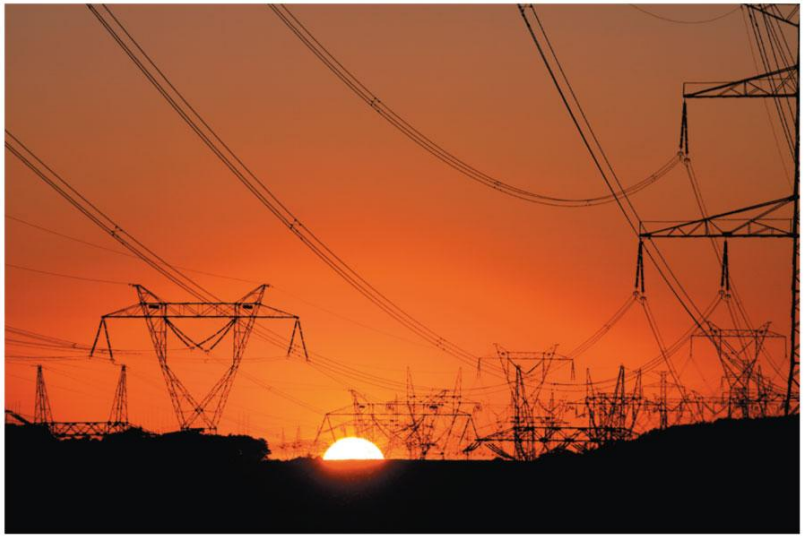
The Speed of a Wave on a String (1 of 2)

- One of the key properties of any wave is the **wave speed**.
- Consider a string in which the tension is F and the linear mass density (mass per unit length) is μ.
- We expect the speed of transverse waves on the string v should increase when the tension F increases, but it should decrease when the mass per unit length μ increases.
- It is shown in your text that the wave speed is:

Speed of a transverse wave on a string $\rightarrow$ $$v = \sqrt{\frac{F}{\mu}}$$ F $\rightarrow$ Tension in string; μ $\rightarrow$ Mass per unit length

The Speed of a Wave on a String (2 of 2)

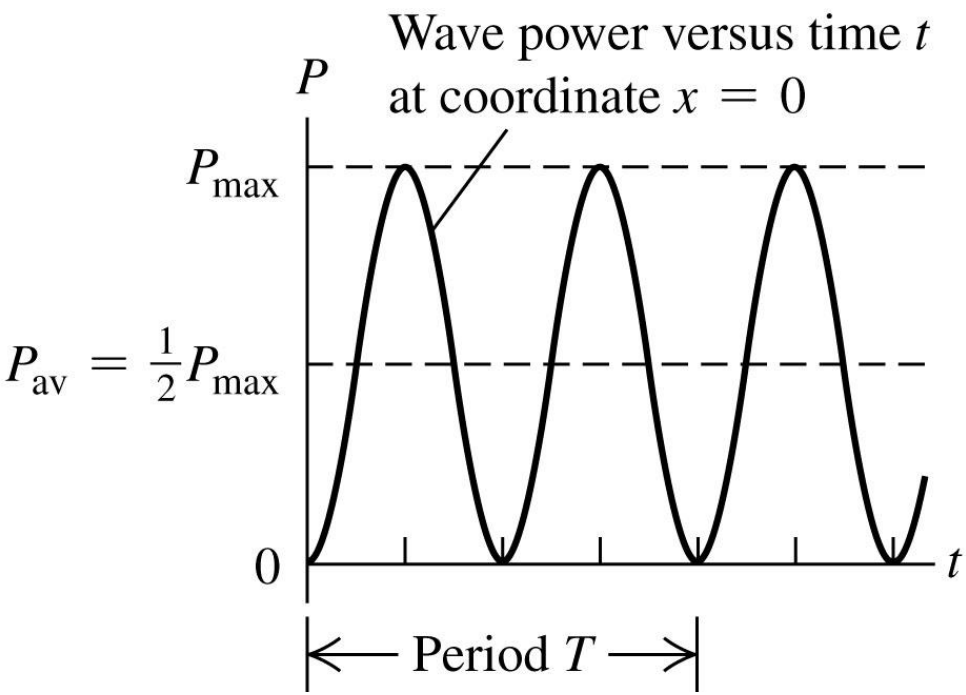
- These transmission cables have a relatively large amount of mass per unit length, and a low tension.
- If the cables are disturbed—say, by a bird landing on them—transverse waves will travel along them at a slow speed.



Ex. 15 One end of a horizontal rope is attached to a prong of an electrically driven tuning fork that vibrates at 120 Hz. The other end passes over a pulley and supports a 1.50-kg mass. The linear mass density of the rope is 0.0480 kg/m. a) What is the speed of a transverse wave on the rope? b) What is the wavelength? c) How would your answers to parts (a) and (b) change if the mass were increase to 3.00 kg?

Power in a Wave (1 of 2)

- Shown is the instantaneous power in a sinusoidal wave.
- The power is never negative, which means that energy never flows opposite to the direction of wave propagation.



Power in a Wave (2 of 2)

- A wave transfers power along a string because it transfers energy.
- The average power is proportional to the **square** of the amplitude and to the **square** of the frequency.
- This result is true for all waves.
- For a transverse wave on a string, the average power is:

Average power, sinusoidal wave on a string

Wave angular frequency

$$P_{\text{av}} = \frac{1}{2}\sqrt{\mu F}\,\omega^2 A^2$$

Wave amplitude

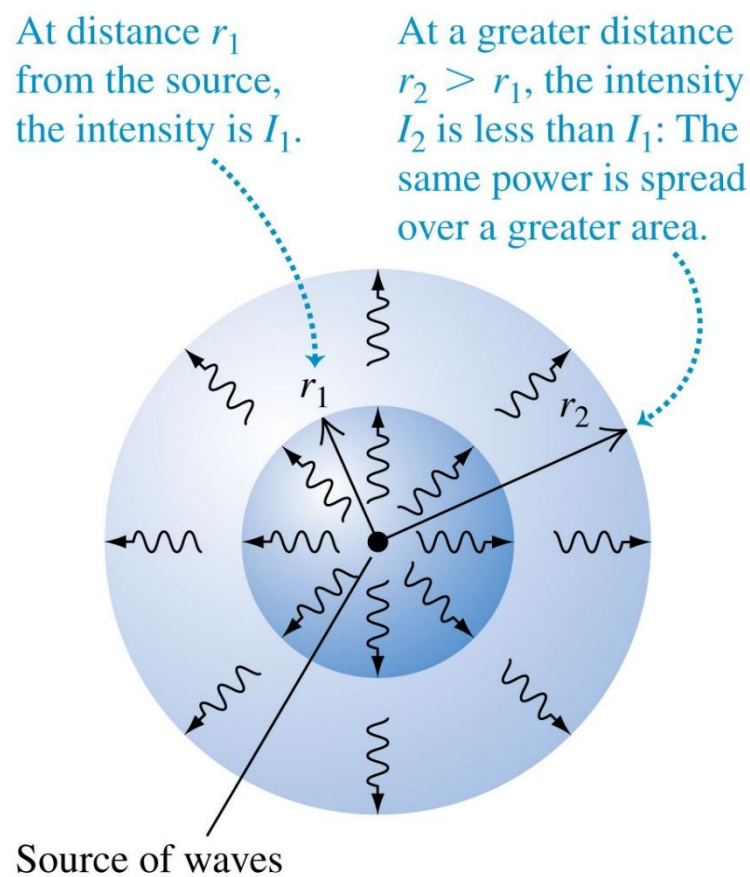
Mass per unit length

Tension in string

Ex. 22 A piano wire with mass 3.00 g and length 80.0 cm is stretched with a tension of 25.0 N. A wave with frequency 120.0 Hz and amplitude 1.6 mm travels along the wire. a) Calculate the average power carried by the wave. b) What happens to the average power if the wave amplitude is halved?

Wave Intensity

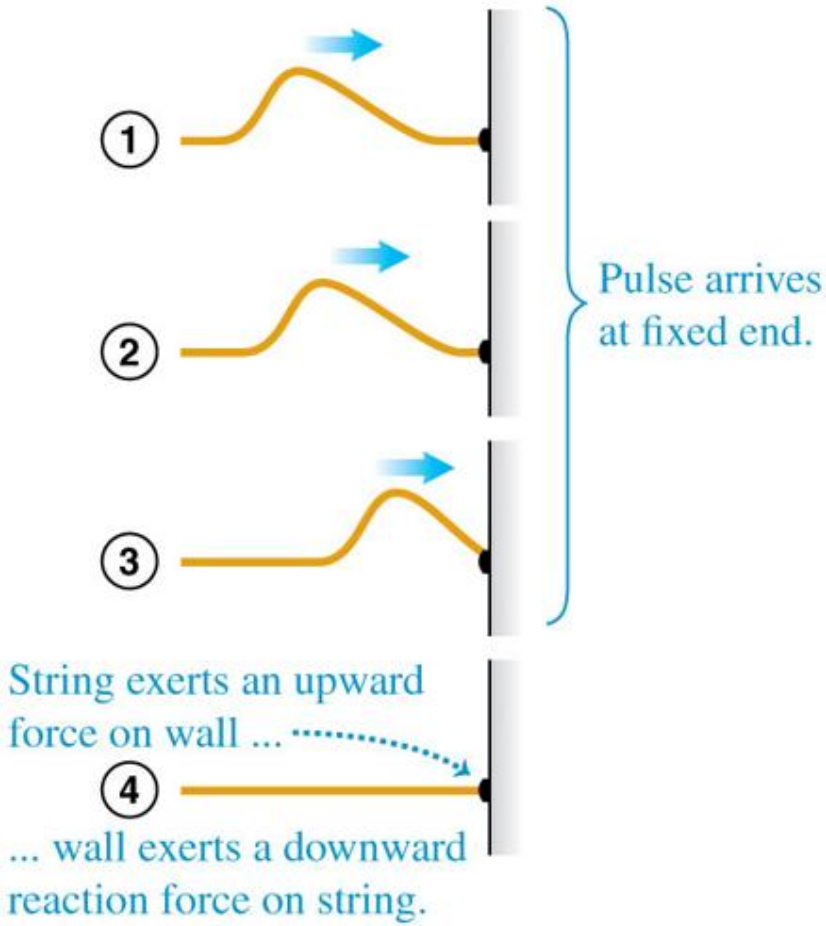
- The **intensity** of a wave is the average power it carries per unit area.
- If the waves spread out uniformly in all directions and no energy is absorbed, the intensity *I* at any distance *r* from a wave source is inversely proportional to r^2.



- Video Tutor Solution: Example 15.5

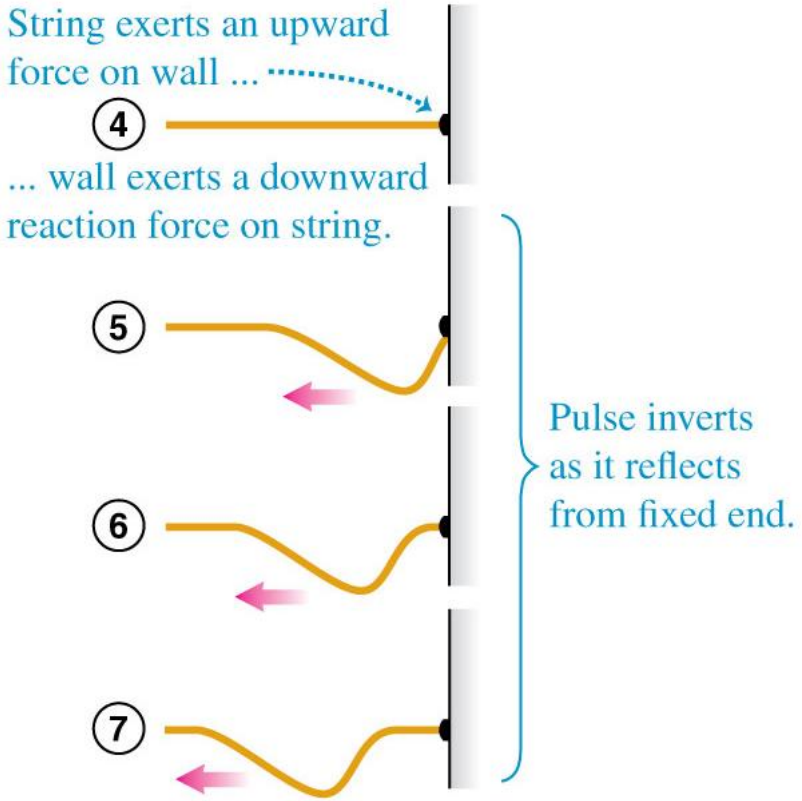
Reflection of a Wave Pulse at a Fixed End of a String (1 of 4)

- What happens when a wave pulse or a sinusoidal wave arrives at the end of the string?
- If the end is fastened to a rigid support, it is a **fixed** end that cannot move.
- The arriving wave exerts a force on the support (drawing 4).



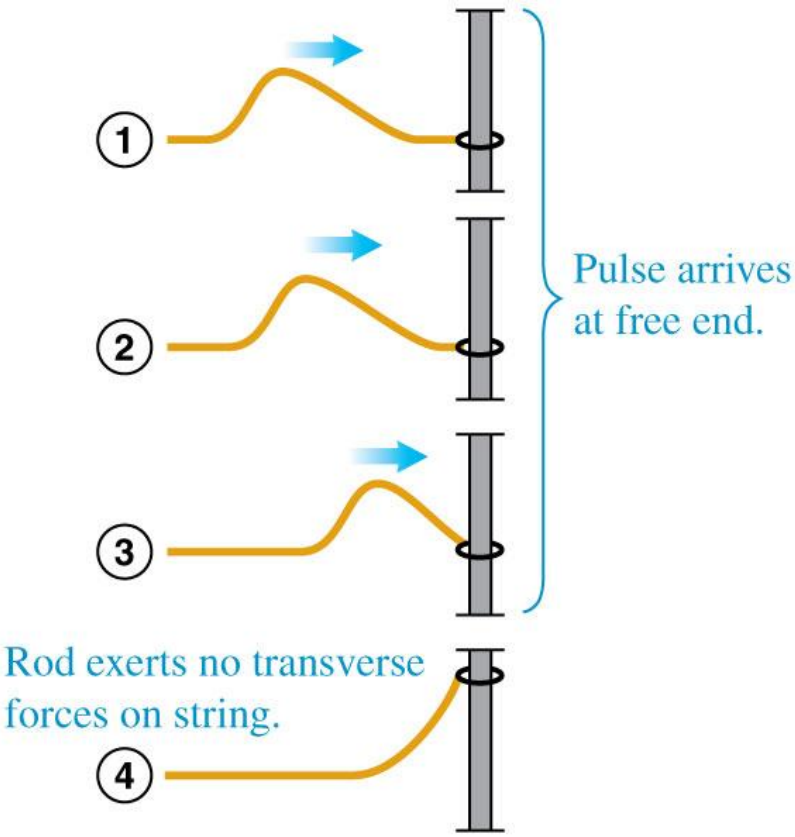
Reflection of a Wave Pulse at a Fixed End of a String (2 of 4)

- The reaction to the force of drawing 4, exerted by the support on the string, “kicks back” on the string and sets up a reflected pulse or wave traveling in the reverse direction.



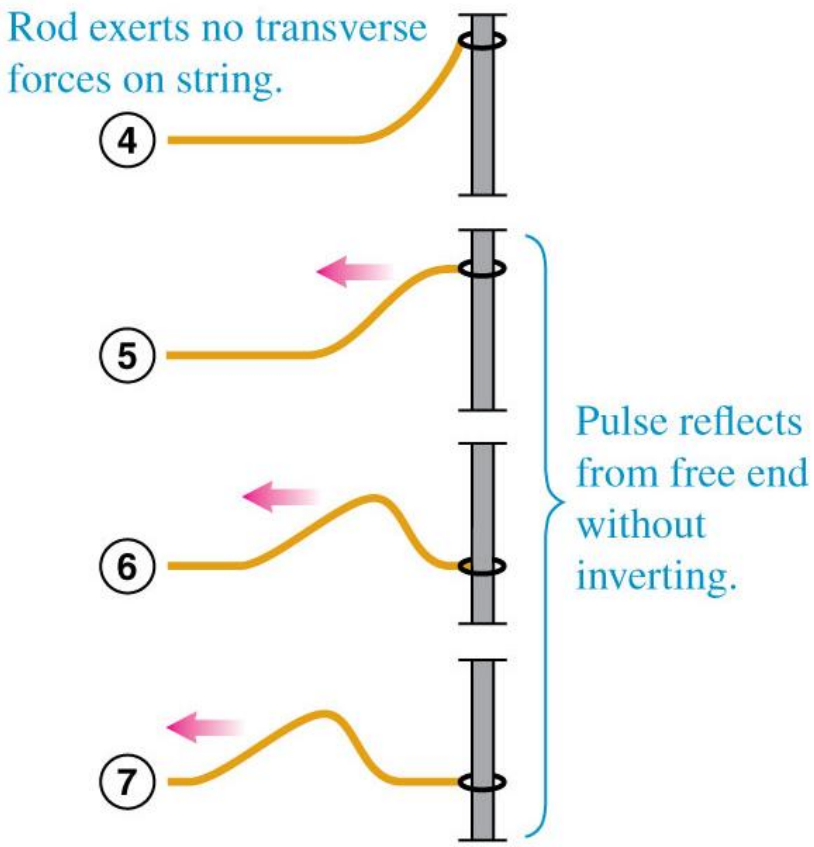
Reflection of a Wave Pulse at a Fixed End of a String (3 of 4)

- A **free end** is one that is perfectly free to move in the direction perpendicular to the length of the string.
- When a wave arrives at this free end, the ring slides along the rod, reaching a maximum displacement, coming momentarily to rest (drawing 4).



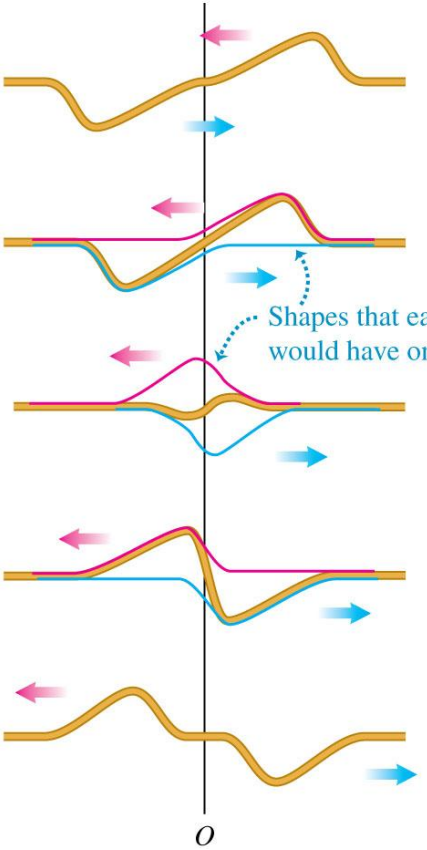
Reflection of a Wave Pulse at a Fixed End of a String (4 of 4)

- In drawing 4, the string is now stretched, giving increased tension, so the free end of the string is pulled back down, and again a reflected pulse is produced.



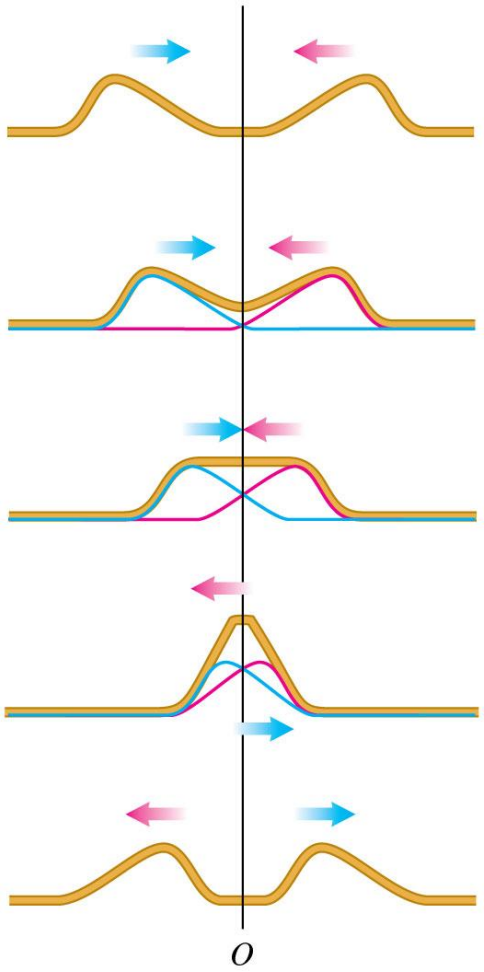
Superposition (1 of 2)

- **Interference** is the result of overlapping waves.
- **Principle of superposition**: When two or more waves overlap, the total displacement is the sum of the displacements of the individual waves.
- Shown is the overlap of two wave pulses—one right side up, one inverted—traveling in opposite directions.
- Time increases from top to bottom.



Superposition (2 of 2)

- Overlap of two wave pulses—both right side up—traveling in opposite directions.
- Time increases from top to bottom.

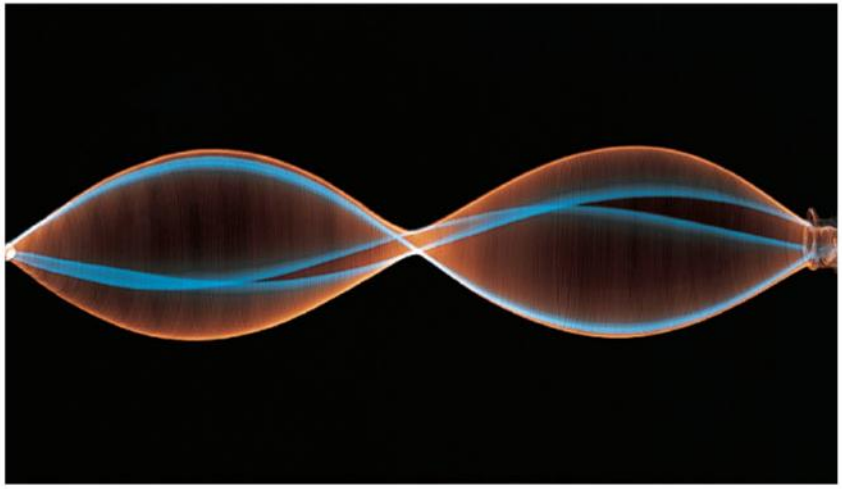


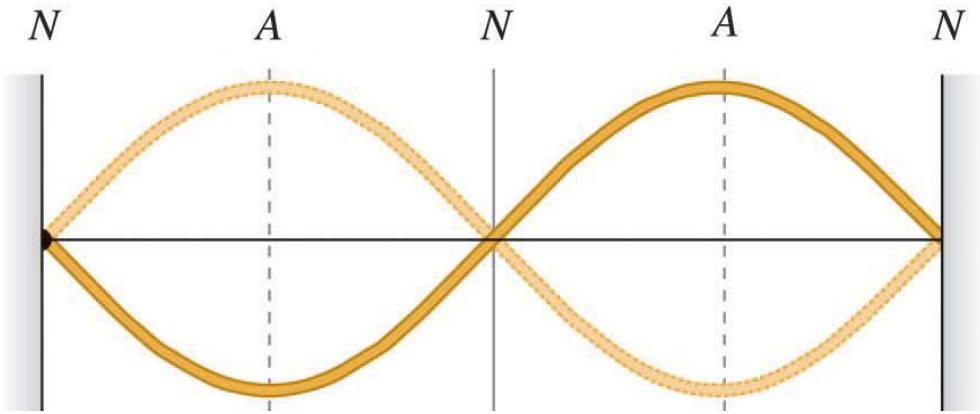
Standing Waves on a String (1 of 3)

- Waves traveling in opposite directions on a taut string interfere with each other.
- The result is a **standing wave** pattern that does not move on the string.
- **Destructive interference** occurs where the wave displacements cancel, and **constructive interference** occurs where the displacements add.
- At the **nodes** no motion occurs, and at the **antinodes** the amplitude of the motion is greatest.
- Video Tutor Demonstration: Out-of-Phase Speakers

Standing Waves on a String (2 of 3)

- This is a time exposure of a standing wave on a string.
- This pattern is called the **second harmonic**.



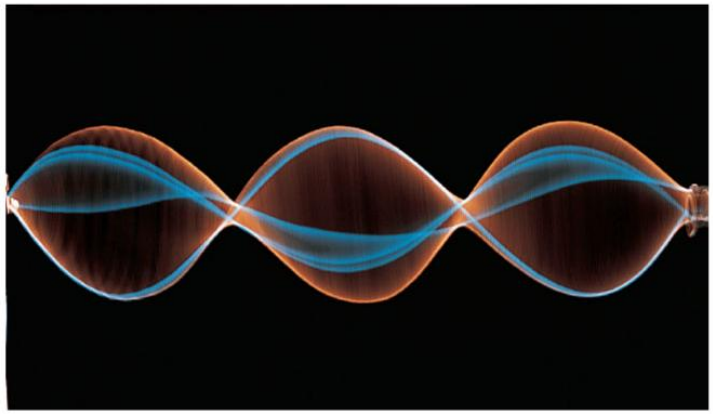


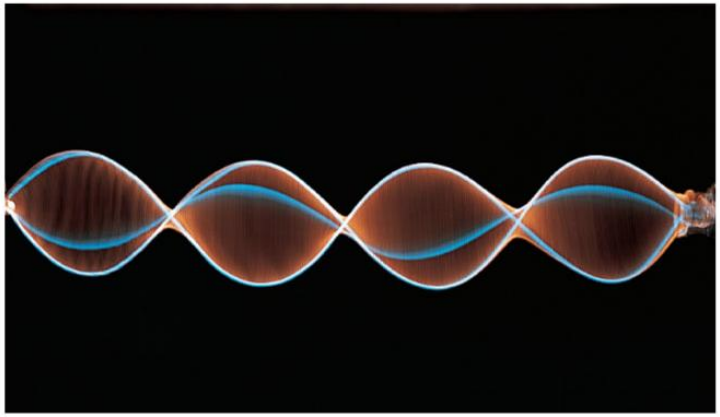
N = **nodes:** points at which the string never moves

A = **antinodes:** points at which the amplitude of string motion is greatest

Standing Waves on a String (3 of 3)

- As the frequency of the oscillation of the right-hand end increases, the pattern of the standing wave changes.
- More nodes and antinodes are present in a higher frequency standing wave.





Ex. 35 Standing waves on a wire are described by Eq. (15.28), with A_{SW}=2.50 mm, $\omega = 942$ rad/s, and $k = 0.750\pi$ rad/m. The left end of the wire is at $x = 0$. At what distances from the left end are (a) the nodes of the standing wave and (b) the antinodes of the standing wave?

The Mathematics of Standing Waves

- We can derive a wave function for the standing wave by adding the wave functions for two waves with equal amplitude, period, and wavelength traveling in opposite directions.
- The wave function for a standing wave on a string in which $x = 0$ is a fixed end is:

Standing wave on a string, fixed end at $x = 0$:

Wave function — Standing-wave amplitude

$$y(x, t) = (A_{SW} \sin kx) \sin \omega t$$

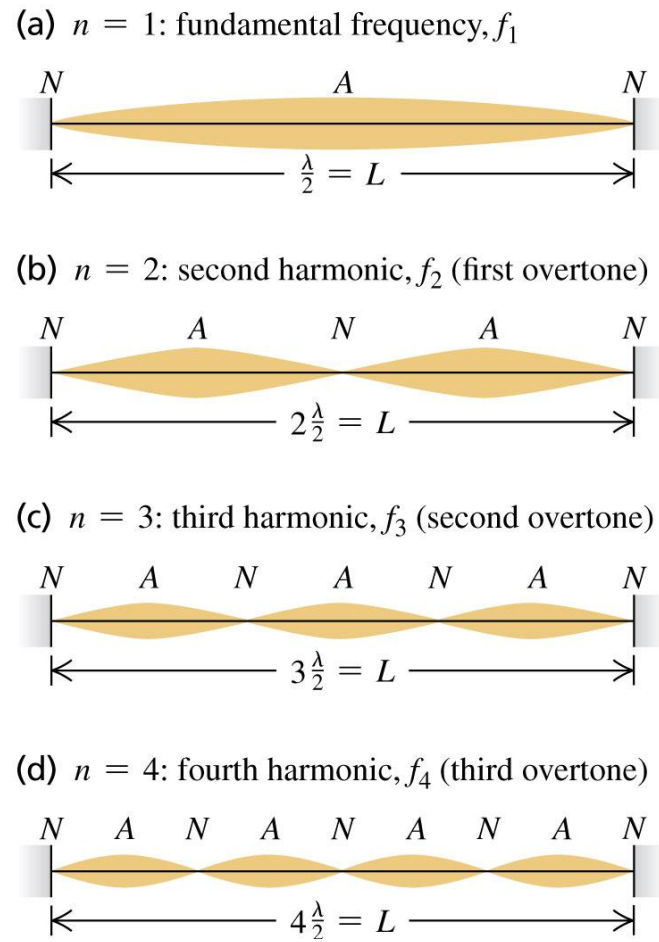
Time

Wave number — Position — Angular frequency

- The standing-wave amplitude A_{SW} is twice the amplitude A of either of the original traveling waves: $A_{SW} = 2A$.

Normal Modes

- For a taut string fixed at both ends, the possible wavelengths are $\lambda_n = \dfrac{2L}{n}$ and the possible frequencies are $f_n = n\dfrac{v}{2L} = nf_1$, where n = 1, 2, 3, …
- f_1 is the **fundamental frequency**, f_2 is the second harmonic (first overtone), f_3 is the third harmonic (second overtone), etc.
- The figure illustrates the first four harmonics.



Standing Waves and String Instruments

- When a string on a musical instrument is plucked, bowed or struck, a standing wave with the fundamental frequency is produced:

Fundamental frequency, string fixed at both ends

$$f_1 = \frac{1}{2L}\sqrt{\frac{F}{\mu}}$$

F: Tension in string

μ: Mass per unit length

L: Length of string

- This is also the frequency of the sound wave created in the surrounding air by the vibrating string.
- Increasing the tension *F* increases the frequency (and the pitch).
- Video Tutor Solution: Example 15.7

Prob. 46 "Not part of the homework." A transverse wave on a rope is given by

$$y(x,t) = (0.75\ \text{cm})\cos\pi\left[\left(0.400\ \text{cm}^{-1}\right)x + \left(250\ \text{s}^{-1}\right)t\right]$$

a) Find the amplitude, period, frequency, wavelength, and speed of propagation. b) Sketch the shape of the rope at these values of t: 0, 0.0005 s, 0.0010 s. c) Is the wave traveling in the $+x$- or $-x$-direction? d) The mass per unit length of the rope is 0.0500 kg/m. Find the tension. e) Find the average power of this wave.

Prob. 69 A large rock that weighs 164.0 N is suspended from the lower end of a thin wire that is 3.00 m long. The density of the rock is 3200 kg/m^3. The mass of the wire is small enough that its effect on the tension in the wire can be ignored. The upper end of the wire is held fixed. When the rock is in air, the fundamental frequency for transverse standing waves on the wire is 42.0 Hz. When the rock is totally submerged in a liquid, with the top of the rock just below the surface, the fundamental frequency for the wire is 28.0 Hz. What is the density of the liquid?